# Earners AI and ML by Aditya Jain sir - Lecture 16

Summary Notes

## For ML-AI : Wednesday to Sunday

## Questions

```
def fun(values):

    total = 0

    for v in values:
        if v % 2 == 0:
            total += v

    return total

numbers = [5, 8, 12, 7, 10]

print(fun(numbers))
```

```
30
```

```
student = {
    "Math": 80,
    "Science": 90,
    "English": 85
}

count = 0

for subject in student:
    if student[subject] >= 85:
        count += 1

print(count)
```

```
#Print the subjects where the marks scored are atleast 85

student = {
    "Math": 80,
    "Science": 90,
    "English": 85
}

sub = []

for v in student:
  if student[v] >= 85:
    sub.append(v)


print(sub)
```

```
['Science', 'English']
```

```
names = ["Rahul", "Amit", "Neha", "Riya"]

answer = ""

for name in names:
    if len(name) == 4:
      print(name, name[-1])
      answer += name[-1]

print(answer)
```

```
Amit t
Neha a
Riya a
taa
```

```
numbers = (10, 15, 20, 25)

result = ()

for n in numbers:
    if n % 5 == 0:
        result += (n // 5,)

print(result)
```

```
(2, 3, 4, 5)
```

```
numbers = (10, 15, 20, 25)

result = ()

for n in numbers:
    if n % 5 == 0:
        result += (n // 5)

print(result)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_900/4284762389.py in <cell line: 0>()
      5 for n in numbers:
      6     if n % 5 == 0:
----> 7         result += (n // 5)
      8
      9 print(result)

TypeError: can only concatenate tuple (not "int") to tuple
```

Next steps: Explain error

```
numbers = (10, 15, 20, 25)

result = (0)

for n in numbers:
    if n % 5 == 0:
        result += (n // 5)

print(result)
```

```
14
```

```
a = (4)
type(a)
```

```
int
```

```
a = (4,)
type(a)
```

```
tuple
```

```
a = [4]
type(a)
```

```
list
```

```
values = {2, 4, 6, 8}

total = 0

for value in values:
    total += value
```

```
print(total // len(values))
```

```
5
```

```
type(values)
```

```
set
```

```
numbers = [3, 6, 9]

new = []

for n in numbers:
    new.append(n * n)

print(max(new))
```

```
81
```

```
student = ("Rahul", 22, 95)

name, age, marks = student

if marks > 90:
    grade = "A"
else:
    grade = "B"

print(name, grade)
```

```
Rahul A
```

```
student = ("Rahul", 22, 95)

name, marks = student
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_900/592252084.py in <cell line: 0>()
      1 student = ("Rahul", 22, 95)
      2
----> 3 name, marks = student

ValueError: too many values to unpack (expected 2)
```

Next steps: Explain error

```
student = ("Rahul", 22, 95)

name, age, marks, grade = student
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
/tmp/ipykernel_900/1377540233.py in <cell line: 0>()
      1 student = ("Rahul", 22, 95)
      2
----> 3 name, age, marks, grade = student

ValueError: not enough values to unpack (expected 4, got 3)
```

Next steps: Explain error

```
fruits = ["Apple", "Banana", "Mango"]

result = []

for fruit in fruits:
    if len(fruit) > 5:
        result.append(fruit.upper())

print(result)
```

```
['BANANA']
```

```
data = {
    "A": 5,
    "B": 10,
    "C": 15
}

total = 0

for key in data:
    total += data[key]

print(total / len(data))
```

```
10.0
```

```
data = {
    "A": 5,
    "B": 10,
    "C": 15
}

total = 0

for key in data:
    total += data[key]

print(total // len(data))
```

```
10
```

```
numbers = [2, 4, 6, 8]

i = 0

total = 0
n = len(numbers)
while i < n:
    total += numbers[i]
    i += 2

print(total)
```

```
8
```

```
text = "Mississippi"

count = {}

for ch in text:
    if ch not in count:
        count[ch] = 1
    else:
        count[ch] += 1

print(count["s"])
```

```
4
```

```
words = ["cat", "elephant", "dog", "tiger"]

longest = words[0]

for word in words:
    if len(word) > len(longest):
        longest = word

print(longest)
```

```
elephant
```

```
students = {
    "Rahul": 90,
```

```
    "Amit": 75,
    "Neha": 95
}

highest_name = ""

highest_marks = 0

for name in students:
    if students[name] > highest_marks:
        highest_marks = students[name]
        highest_name = name

print(highest_name)
```

```
Neha
```

## Set

```
students = [
    "Rahul",
    "Amit",
    "Rahul",
    "Neha",
    "Amit",
    "Rahul"
]

print(students)
```

```
['Rahul', 'Amit', 'Rahul', 'Neha', 'Amit', 'Rahul']
```

```
type(students)
```

```
list
```

```
#Write a code to only store the unique occurances of the elements in the list.

students = [
    "Rahul",
    "Amit",
    "Rahul",
    "Neha",
    "Amit",
    "Rahul"
]

s1 = []

for student in students:
  if student not in s1:
    s1.append(student)


print(s1)
```

```
['Rahul', 'Amit', 'Neha']
```

```
# Set:

# 1. Stores only the unique values
# 2. Unordered
# 3. Mutable (add/remove elements)
```

```
num = {3,6,8,2,1}
type(num)
```

```
set
```

```
fruits = {"Apple", "Banana", "Orange"}
len(fruits)
```

```
3
```

```
num = {4, 5, 7, 2, 4, 8, 1}
len(num)
```

```
6
```

```
s = {4, 5.5,"AJ", False}
type(s)
```

```
set
```

```
l = []
t = ()
d = {}
```

```
type(d)
```

```
dict
```

```
s = set()
type(s)
```

```
set
```

```
len(s)
```

```
0
```

```
num = {4, 5, 7, 2, 4, 8, 1}
print(num)
```

```
{1, 2, 4, 5, 7, 8}
```

```
num[0]
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_900/3703492311.py in <cell line: 0>()
----> 1 num[0]

TypeError: 'set' object is not subscriptable
```

Next steps: Explain error

```
for n in num:
  print(n)
```

```
1
2
4
5
7
8
```

```
num.add(10)
print(num)
```

```
{1, 2, 4, 5, 7, 8, 10}
```

```
num.remove(4)
print(num)
```

```
{1, 2, 5, 7, 8, 10}
```

```
num.remove(4)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_900/1514787385.py in <cell line: 0>()
----> 1 num.remove(4)

KeyError: 4
```

Next steps: Explain error

```
print(num)
```

```
{1, 2, 5, 7, 8, 10}
```

```
item = num.pop()
print(item)
```

```
5
```

```
print(num)
```

```
{7, 8, 10}
```

```
num.clear()
```

```
print(num)
```

```
set()
```

```
fruits
```

```
{'Apple', 'Banana', 'Orange'}
```

```
"Orange" in fruits
```

```
True
```

```
"Grapes" in fruits
```

```
False
```

```
#union

A = {1,2,3}

B = {3,4,5}

print(A | B)
```

```
{1, 2, 3, 4, 5}
```

```
#intersection
A = {1,2,3}

B = {3,4,5}

print(A & B)
```

```
{3}
```

```
#set difference

A = {1,2,3}

B = {3,4,5}

print(A - B)
```

```
{1, 2}
```

## Quiz

```
numbers = {10, 20, 30, 20, 10}

total = 0

for num in numbers:
    total += num

print(total)
```

```
60
```

```
fruits = {"Apple", "Banana"}

fruits.add("Mango")
fruits.add("Apple")

print(len(fruits))
```

```
3
```

```
numbers = {5, 10, 15}

numbers.remove(10)

print(sum(numbers))
```

```
20
```

```
colors = {"Red", "Green", "Blue"}

count = 0

for color in colors:
    if "e" in color.lower():
        count += 1

print(count)
```

```
3
```

```
numbers = {2, 4, 6, 8}

largest = max(numbers)

smallest = min(numbers)

print(largest - smallest)
```

```
6
```

```
numbers = [10, 20, 20, 30, 10]

unique = set(numbers)

print(len(unique))
```

```
3
```

```
fruits = {"Apple", "Banana"}

if "Mango" in fruits:
    print("Yes")
else:
    print("No")
```

```
No
```

```
cities = {"Delhi", "Mumbai"}

cities.add("Pune")
```

```
cities.remove("Delhi")

print(len(cities))
```

```
2
```

```
text = "banana"

letters = set(text)

print(len(letters))
```

```
3
```

```
A = {"Python", "SQL"}

B = {"Python", "Power BI"}

print(len(A & B))
```

```
1
```

```
print(A&B)
```

```
{'Python'}
```

```
len({'Python'})
```

```
1
```

```
numbers = {5, 10, 15, 20}

count = 0

for num in numbers:
    if num > 10:
        count += 1

print(count)
```

```
2
```

## HW

Double-click (or enter) to edit

```
words = ["apple", "banana", "apple", "mango", "banana"]

unique = set(words)

result = ""

for word in unique:
    result += word[0]

print(len(result))
```

```
text = "mississippi"

letters = set(text)

count = 0

for ch in letters:
    if ch in "aeiou":
        count += 1

print(count)
```

```
A = {10, 20, 30}

B = {30, 40, 50}

C = A.union(B)

print(min(C), max(C))
```

```
numbers = [1, 2, 2, 3, 4, 4, 5]

unique = set(numbers)

even = set()

for num in unique:
    if num % 2 == 0:
        even.add(num)

print(len(even))
```

```
students = {
    "Rahul",
    "Amit",
    "Neha"
}

students.remove("Amit")

students.add("Riya")

print(len(students))
```

```
colors = {"Red", "Green", "Blue"}

result = set()

for color in colors:
    result.add(color.upper())

print(len(result))
```

```
numbers = [10, 20, 20, 30, 40, 40]

A = set(numbers)

B = {20, 50}

result = A - B

print(sum(result))
```